PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Effectiveness of Case-based Learning in Chinese Dental Education: A systematic review and meta-analysis
AUTHORS	Dong, Haitao; Guo, Chunlan; Zhou, Lian; Zhao, Jizhi; Wu, Xiaomin; Zhang, Xinyuan; Zhang, Xin

VERSION 1 – REVIEW

REVIEWER	Yang, Guoli Zhejiang University School of Medicine
REVIEW RETURNED	07-May-2021

GENERAL COMMENTS	This is a comprehensive and informative review of how CBL practice has improved dental teaching outcomes in China. The design of the study is appropriate and well finished. The analyses and discussion of the studies are objective and the limitations are adequately discussed. However, there are some minor grammar mistakes and language inaccuracies that might cause some trouble in understanding.(e.g. line 36 page 4, 'goes on like students'/ line 9 page 5,'subsequently increasing their success rates.'). A small amount of refining job could be done.

REVIEWER	Shrestha, Ashish BP Koirala Institute of Health Sciences
REVIEW RETURNED	27-May-2021

GENERAL COMMENTS	Very well compiled scientific information to justify the application of CBL in the dental curriculum in China. Best wishes.

REVIEWER	Jum'ah, Ahmad A Jordan University of Science and Technology, Faculty of Dentistry
REVIEW RETURNED	11-Nov-2021

GENERAL COMMENTS	Thank you for the opportunity to review this manuscript. This is a timely and interesting systematic review with meta-analysis that investigated the effectiveness of CBL in Chinese dental educational systems. The manuscript is concise and well-structured. Nevertheless, there are some deficiencies that I have stated below: - There are significant flaws related to the English language that require serious revision. - Abstract-methods: the word literatures may be inappropriate - Introduction: There are several lengthy statements that hinder smooth flow. - Several statements start with And or numeral?

	- Methods-Selection criteria: how it was determined that the RCT had a sufficient sample size? - Methods: Teaching satisfaction: Despite being self-explainatory, objectively defining satisfactory/general/poor teaching effects might be required! - Discussion: Might be helpful to compare your data with previously published reviews

REVIEWER	Heumann, Christian Ludwig-Maximilians-Universitat Munchen, Faculty of Mathematics, Informatics and Statistics, Department of Statistics
REVIEW RETURNED	02-Dec-2021

GENERAL COMMENTS	I congratulate the authors to their well and clearly written paper. I have two comments which should be addressed in the revision: 1) There is an error in the mean and the confidence interval of the knowledge scores, Figure 4 gives probably the correct values (SMD 1.58, CI [0.95 , 2.20]). The error appears in the abstract (page 3, line 19/20) and in the Results section (page 9, line 9), 2) In the statistical analysis section (page 7, line 17/18) I will drop the sentence "If the chi-squared [...]". First of all, the fixed effects model was never used in the analysis (only the random effects model which is suitable in that context in my point of view). Secondly, to make the selection of a model (fixed vs. random effects) dependent on a pre-test has bad statistical properties.

VERSION 1 – AUTHOR RESPONSE

Reviewer Comments:
Reviewer #1
This is a comprehensive and informative review of how CBL practice has improved dental teaching outcomes in China. The design of the study is appropriate and well finished. The analyses and discussion of the studies are objective and the limitations are adequately discussed. However, there are some minor grammar mistakes and language inaccuracies that might cause some trouble in understanding. (e.g. line 36 page 4, 'goes on like students'/ line 9 page 5,'subsequently increasing their success rates.'). A small amount of refining job could be done.

Answer：Thank you so much for this comment and suggestion. We had asked a native English collaborator to check and revised the grammar of our manuscript, the following changes were made:
Page 4, line 33: "how to" was changed to "to".
Page 4, line 35: The sentence "In the CBL method, the student-teacher relationship goes on like students take on the learning responsibilities by analyzing the given case, while the teacher's role is a prompter and guide helping students justify their analyses in designing effective treatment strategy" was re-written to "So, the student-teacher relationship goes on like that, the students take on the learning responsibilities by analyzing the given case and designing effective treatment strategy, while the teacher's role is a prompter and guide to help students justify their analyses."
Page 5, line 2: "universities" was changed to "schools".
Page 5, line 5: "model" was changed to "models".
Page 5, line 5: "several dental" was changed to "several more dental".
Page 5, line 9: "subsequently increasing their success rates." was changed to "subsequently presenting its increasing success rates. "

2. Reviewer# 2
Very well compiled scientific information to justify the application of CBL in the dental curriculum in China. Best wishes.
Answer：Thank you so much for your positive comment.

3. Reviewer# 3 Thank you for the opportunity to review this manuscript. This is a timely and interesting systematic review with meta-analysis that investigated the effectiveness of CBL in Chinese dental educational systems. The manuscript is concise and well-structured. Nevertheless, there are some deficiencies that I have stated below: - There are significant flaws related to the English language that require serious revision. - Abstract-methods: the word literatures may be inappropriate - Introduction: There are several lengthy statements that hinder smooth flow. - Several statements start with And or numeral? - Methods-Selection criteria: how it was determined that the RCT had a sufficient sample size? - Methods: Teaching satisfaction: Despite being self-explanatory, objectively defining satisfactory/general/poor teaching effects might be required! - Discussion: Might be helpful to compare your data with previously published reviews

(1) There are significant flaws related to the English language that require serious revision
Answer: Thank you so much for this suggestion. We had asked a native English collaborator to check and revised the grammar of our manuscript, please refer to the Answers to the Reviewer 1.

(2) Abstract-methods: the word literatures may be inappropriate
Answer: Thank you for your valuable suggestion, we had revised the word
Page 3, line 14: "Literatures" was changed to "Studies"

(3) Introduction: There are several lengthy statements that hinder smooth flow.
Answer: Thank you for your valuable suggestion, we had revised the sentences:
Page 4, line 33: The sentence "CBL methods generally focus on using wide ranges of medical and clinical cases to teach students about actual patient care situations by guiding students on how to implement their acquired knowledge foundation in making decisions on problems that they may encounter in practice." was re-written to “CBL methods generally focus on using wide ranges of medical and clinical cases to teach students about actual patient care situations. In these curriculum, the teachers mainly guide the students to implement their acquired knowledge foundation in making decisions on problems that they may encounter in practice."
Page 4, line 40: The sentence "helping students justify their analyses in designing effective treatment strategy" was re-written to "by analyzing the given case and designing effective treatment strategy, while the teacher’s role is a prompter and guide to help students justify their analyses"
Page 5, line 5: The statement "……by including the most effective teaching concepts which were already being applied in diverse educational strategies, like CBL pedagogy" was changed to "These models included the most effective teaching concepts and were already being applied in diverse educational strategies, like CBL pedagogy."

(4) Several statements start with And or numeral?
Answer: Thank you so much for this comments, we had revised accordingly.
Page 4, line 14: "And" was deleted.
Page 9, line 28: "8" was changed to "Eight"
Page 9, line 41: "11" was changed to "Eleven"

(5) Methods-Selection criteria: how it was determined that the RCT had a sufficient sample size?
Answer: Thank you very much for this comment. In all the including studies, the results were reliable, which approved that the studies have sufficient sample size. In order to avoid ambiguity,
"(2) sufficient sample size was included in each RCTs; " was dropped on Page 6, line 11/12.

(6) Methods: Teaching satisfaction: Despite being self-explanatory, objectively defining satisfactory/general/poor teaching effects might be required!
Answer: Thank you so much for your constructive suggestion, we had added the objectively defining accordingly:
Page 6, Line 57: "In the including studies, the teaching satisfaction was usually evaluated by Likert scale. From 5 to 1, the decrease of the scores mean the decline of satisfaction. When the scores were 5~4, teaching satisfaction was defined as significant. When the scores were 3~2, teaching satisfaction was defined as general. When the scores were 1, teaching satisfaction was defined as poor." was added.

(7) Discussion: Might be helpful to compare your data with previously published reviews
Answer: Thank you so much for your valuable suggestion, we had added some comparison with other review accordingly:
Page 11, Line 54: "As we know, this is the first systematic review and meta-analysis of CBL practice in Chinese dental education. Compared with other reviews on teaching methods reform in Chinese dental education previously, such as meta-analysis on PBL practice in Chinese dental education published by Huang et al [40], the present study had two strengths. Firstly, a subgroup analysis was performed comparing the course type between the theoretical courses and the practical courses. Secondly, this study independently used funnel plots to assess publication bias. The plots were generally symmetrical and showed a lower publication bias. " was added.

4. Reviewer #4
I congratulate the authors to their well and clearly written paper. I have two comments which should be addressed in the revision:
(1) There is an error in the mean and the confidence interval of the knowledge scores, Figure 4 gives probably the correct values (SMD 1.58, CI [0.95, 2.20]). The error appears in the abstract (page 3, line 19/20) and in the Results section (page 9, line 9).
Answer: Thank you for your careful review and valuable suggestion, the following changes were made accordingly:
Page 3, Line 19/20: "SMD=1.53, CI [0.93-2.53] " was changed to "SMD 1.58, CI [0.95, 2.20] "
Page 9, Line 9: "SMD=1.53, CI [0.93-2.53] " was changed to "SMD 1.58, CI [0.95, 2.20] "

(2) In the statistical analysis section (page 7, line 17/18) I will drop the sentence "If the chi-squared [...]". First of all, the fixed effects model was never used in the analysis (only the random effects model which is suitable in that context in my point of view). Secondly, to make the selection of a model (fixed vs. random effects) dependent on a pre-test has bad statistical properties.
Answer: Thank you for your constructive suggestion, we had revised accordingly:
Page 7, line 17/18: The sentence "The chi-squared test and I2 statistic were used to assess statistical heterogeneity of the data. If the chi-squared test >0.10 or if the I2 <50%, the fixed-effects model was chosen to pool data if there was no heterogeneity. Otherwise, a random-effects model was adopted." was deleted.